

Week 1

Assignment

CMI MSc DS Crash Course

Cheenta School of Statistics and Data Science

The solutions will be discussed live in the classes.

Problem 1

If $\log_{12} 27 = \lambda$ then $\log_6 16 =$

- (A) $\frac{1+\lambda}{\lambda}$
- (B) $\frac{2\lambda}{3-\lambda}$
- (C) $5 \left(\frac{2+\lambda}{2-\lambda} \right)$
- (D) $4 \left(\frac{3-\lambda}{3+\lambda} \right)$

Problem 2

If $x \neq 1$ and $(\log_3 x)(\log_4 x)(\log_5 x) = (\log_4 x \log_5 x) + (\log_3 x \log_5 x) + (\log_4 x \log_3 x)$ then the value of x is-

- (A) 20
- (B) 30
- (C) 10
- (D) 60

Problem 3

If $\log_{\sqrt{3}} x + \log_{\sqrt[4]{3}} x + \log_{\sqrt[6]{3}} x + \cdots + \log_{16\sqrt{3}} x = 36$ then the value of x is-

(A) 3

(B) $\sqrt{3}$ (C) $4\sqrt{3}$

(D) 9

Problem 4

$\log_1 x \quad \log_1 y \quad \log_1 z$ If $\frac{\frac{1}{2}}{b-c} = \frac{\frac{1}{2}}{c-a} = \frac{\frac{1}{2}}{b-c}$ then show that, $x^a y^b z^c = xyz$.

Problem 5

Find the values : 1) $\log_6 \sqrt{6\sqrt{6\sqrt{6\cdots\infty}}}$

Problem 6

Show that, $\log_a x^{\frac{1}{\log_{ab} x}}$
 $= 1 + \log_a b$.

Problem 7

If $x = e^y - e^{-y} \frac{1}{e^y + e^{-y}}$

show that, $y = \frac{1}{2 \log_e \frac{1+x}{1-x}}$.

Problem 8

Solve :

$$\log_5 \left(5^{\frac{1}{x}} + 125 \right) = \log_5 6 + 1 + \frac{1}{2x}$$

Problem 9

$$x + \log_{10} (1 + 2^x) = x \log_{10} 5 + \log_{10} 6$$

Problem 10

If $x = \log_a(bc)$, $y = \log_b(ca)$ and $z = \log_c(ab)$ show that,

$$(i) \frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1$$

$$(ii) x + y + z + 2 = xyz.$$

Problem 11

If $\frac{x(y+z-x)}{\log x} = \frac{y(z+x-y)}{\log y} = \frac{z(x+y-z)}{\log z}$ then show that $y^z \cdot z^y = x^y \cdot y^x = z^x \cdot x^z$

Problem 12

$$\text{Solve : } 4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}$$

Problem 13

If $(ax)^{\log a} = (by)^{\log b}$ and $b^{\log x} = a^{\log y}$ then show that $x = \frac{1}{a}, y = \frac{1}{b}$.

Problem 14

If $\log(x + y + z) = \log x + \log y + \log z$, then prove that,

$$(i) \log \left(\frac{2x}{1-x^2} + \frac{2y}{1-y^2} + \frac{2z}{1-z^2} \right) = \log \left(\frac{2x}{1-x^2} \right) + \log \left(\frac{2y}{1-y^2} \right) + \log \left(\frac{2z}{1-z^2} \right)$$

$$(ii) \log \left(\frac{3x-x^3}{1-3x^2} + \frac{3y-y^3}{1-3y^2} + \frac{3z-z^3}{1-3z^2} \right) = \log \left(\frac{3x-x^3}{1-3x^2} \right) + \log \left(\frac{3y-y^3}{1-3y^2} \right) + \log \left(\frac{3z-z^3}{1-3z^2} \right)$$

Problem 15

Evaluate the following using L' Hospital rule.

$$\lim_{x \rightarrow 1} \frac{1 + \log x - x}{1 - 2x + x^2}$$